

ESTHERVILLE, IA, USA

WMO: 726499

Lat: 43.401N

Lon: 94.747W

Elev: 401

StdP: 96.59

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 94971

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.0	-21.7	-27.4	0.3	-23.7	-25.0	0.4	-21.3	15.5	-8.0	14.4	-8.5	4.5	310	0.652

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	32.1	23.6	30.6	22.6	28.9	21.6	25.4	29.7	24.3	28.5	23.3	27.4	6.8	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.9	28.0	22.9	18.5	27.0	22.1	17.6	26.1	80.3	29.6	75.6	28.4	71.3	27.1	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.4	11.9	10.8	DB	-28.0	34.9	3.2	1.5	-30.3	36.0	-32.2	36.9	-34.0	37.7	-36.3	38.9	
			WB	-28.1	27.4	3.1	1.0	-30.3	28.1	-32.1	28.7	-33.9	29.2	-36.1	30.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	-8.9	-6.8	0.4	8.0	14.7	20.2	22.2	20.3	16.8	9.2	1.3	-6.1
	DBStd	12.23	7.21	7.14	6.92	5.56	4.97	3.65	3.15	2.99	4.51	5.37	6.11	6.65
	HDD10.0	2326	585	470	307	101	14	0	0	0	4	79	266	499
	HDD18.3	4227	843	703	558	312	134	22	6	14	82	286	509	758
	CDD10.0	1482	0	0	8	41	160	306	378	320	207	56	6	0
	CDD18.3	342	0	0	0	2	22	78	126	76	35	4	0	0
	CDH23.3	3203	0	0	3	41	264	744	1156	593	343	59	1	0
	CDH26.7	964	0	0	0	11	84	246	377	141	92	14	0	0
Wind	WSAvg	5.2	5.9	5.9	5.9	6.3	6.0	4.9	3.8	3.4	4.2	5.0	5.6	5.7
Precipitation	PrecAvg	710	16	15	43	75	95	112	92	88	80	51	32	15
	PrecMax	1144	75	87	103	195	237	343	267	214	245	212	136	50
	PrecMin	471	0	0	0	16	24	36	8	4	6	2	0	0
	PrecStd	156	19	19	28	42	51	64	56	47	56	43	28	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.0	12.5	22.2	27.9	32.4	33.9	33.8	32.2	32.0	28.6	21.2	12.2
		MCWB	4.2	9.0	14.2	16.6	20.1	22.8	25.5	25.2	22.1	17.9	13.0	9.4
	2%	DB	4.0	7.8	17.3	23.5	28.9	31.5	32.1	29.9	29.1	24.2	17.0	7.8
		MCWB	1.8	5.7	11.9	14.3	18.4	22.2	24.9	23.9	21.1	16.2	11.5	5.0
	5%	DB	2.2	4.2	13.6	20.5	26.3	29.5	30.4	28.0	27.3	21.3	13.1	4.1
		MCWB	0.9	2.7	9.9	12.4	17.1	20.7	23.9	22.7	19.7	15.0	8.6	2.0
	10%	DB	0.6	2.4	10.0	17.5	23.6	27.6	28.8	26.9	25.1	18.2	10.2	2.2
		MCWB	-0.2	1.2	7.4	11.4	16.0	19.8	22.7	21.7	18.7	13.1	7.0	1.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.8	9.7	16.1	17.4	22.4	25.1	27.6	26.8	24.0	20.9	14.6	10.5
		MCDB	8.1	12.9	18.8	23.8	29.0	30.9	31.3	30.5	28.9	26.0	19.2	12.0
	2%	WB	2.2	5.7	12.9	15.5	20.3	23.7	25.9	24.9	22.6	18.0	11.9	5.1
		MCDB	3.8	7.9	16.4	21.4	26.2	29.1	30.0	28.5	26.9	21.7	15.8	7.7
	5%	WB	0.9	3.0	10.2	13.8	18.8	22.5	24.9	23.7	21.4	15.9	9.4	2.6
		MCDB	2.0	4.3	13.3	19.2	24.5	27.5	29.1	27.1	25.4	20.0	12.8	4.2
	10%	WB	-0.3	1.3	7.2	12.0	17.2	21.2	23.8	22.4	20.1	13.6	7.0	0.9
		MCDB	0.5	2.2	10.1	16.2	21.4	25.9	27.9	25.6	23.7	18.1	10.1	1.9
Mean Daily Temperature Range	5% DB	MDBR	9.0	9.2	9.4	11.9	11.8	11.2	11.1	10.9	12.8	12.1	10.2	8.5
		MCDBR	10.6	10.4	15.0	17.7	15.8	13.7	11.9	12.0	14.7	16.4	15.3	11.8
		MCWBR	8.9	8.4	9.7	9.7	7.6	6.2	6.4	6.9	7.3	9.1	9.8	9.4
	5% WB	MCDBR	9.7	9.8	13.9	15.0	12.7	11.4	10.9	10.8	12.1	13.5	13.9	10.8
		MCWBR	8.3	8.0	9.4	9.1	7.0	6.3	6.6	6.8	7.0	8.7	9.6	8.8
Clear-Sky Solar Irradiance	taub	0.271	0.282	0.310	0.360	0.383	0.401	0.416	0.407	0.365	0.327	0.286	0.266	
	taud	2.435	2.420	2.446	2.336	2.343	2.344	2.283	2.292	2.381	2.480	2.521	2.475	
	Ebn at Noon	875	924	935	904	889	871	852	848	861	858	852	849	
	Edh at Noon	73	88	99	120	123	123	129	124	105	83	67	64	
All-Sky Solar Radiation	RadAvg	1.85	2.92	3.82	4.71	5.52	6.11	6.52	5.42	4.37	2.91	1.96	1.44	
	RadStd	0.20	0.31	0.37	0.44	0.39	0.46	0.28	0.39	0.29	0.38	0.15	0.13	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (56 neighbors)										
		-1.04	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page